

1. Introduction to Ethical Hacking

Ethical hacking is the authorized practice of bypassing system security to identify potential data breaches and threats in a network or system. Ethical hackers use the same tools and techniques as malicious hackers but do so legally and with permission to help strengthen security.

2. Creating a Security Evaluation Plan

Before any ethical hacking, a plan is created outlining:

- Scope of the test (what systems to test)
- Rules of engagement (how tests are conducted)
- Timeframes
- Reporting requirements
The plan ensures the testing is controlled and legal.

3. Types of Ethical Hacks

- **Black Box Testing:** No prior knowledge of the system; simulates an external attacker.
- **White Box Testing:** Full knowledge of system internals; simulates insider threat or thorough assessment.
- **Gray Box Testing:** Partial knowledge; combines both approaches.

4. Footprinting and Social Engineering

- **Footprinting:** Gathering information about a target system/network using public resources (DNS records, IP addresses, employee info).
- **Social Engineering:** Manipulating people to reveal confidential info (phishing, pretexting).

5. Traceroute in Footprinting

Traceroute is a network diagnostic tool that tracks the path packets take to reach a host. It helps identify intermediate devices and network topology, useful in footprinting to map a target's network.

6. Port Scanning, Network Scanning, Vulnerability Scanning

- **Port Scanning:** Identifies open ports on a system that can be entry points.
- **Network Scanning:** Discovers active devices on a network.
- **Vulnerability Scanning:** Detects security weaknesses on hosts or networks.

7. Scanning Methodologies

- **Active Scanning:** Directly interacts with the target (e.g., sending packets).
- **Passive Scanning:** Observes network traffic without direct interaction.
- **Stealth Scanning:** Avoids detection by firewalls or IDS.

8. Common Scan Types

- **SYN Scan (Half-open):** Sends SYN packet to open TCP connection but does not complete handshake, less likely to be logged.
- **Stealth Scan:** Attempts to avoid detection by firewalls or IDS, often using partial connections.
- **XMAS Scan:** Sends packets with FIN, URG, and PSH flags set; can reveal open/closed ports based on responses.
- **NULL Scan:** Sends packets with no flags set; helps identify open ports by how system responds.
- **IDLE Scan:** Uses a "zombie" host to send packets, hiding the attacker's IP.
- **FIN Scan:** Sends FIN packets to close connections, used to probe ports stealthily.